UNIVERSITÀ DEGLI STUDI DI MILANO

Master’s Program in Informatics

Project Report – Natural Language Processing

A Semantic and Topic-Based Analysis of Research Trends Using NLP Techniques

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1. Introduction

Natural Language Processing (NLP) research has expanded rapidly over the last decade, resulting in a growing diversity of research topics, methodologies, and applications. As journals continue to publish a large number of papers, evaluating whether published articles remain aligned with the journal's thematic focus becomes increasingly important. Traditional evaluation methods often rely on manual review, which can be time-consuming and subjective.

This project aims to analyze thematic alignment in NLP journal articles using Natural Language Processing techniques. By applying semantic embeddings and similarity measures, the study evaluates how closely research papers align with the core themes of the journal over time. In addition to alignment analysis, the project investigates publication trends, semantic consistency, topic evolution, and thematic outliers.

To achieve this, article abstracts and metadata were collected and processed using NLP methods. Sentence-BERT embeddings were used to represent semantic meaning, cosine similarity was applied to calculate thematic alignment, and TF-IDF-based topic extraction and semantic grouping were used for topic analysis and trend exploration. The results provide insights into how research themes evolve and how consistent papers remain within the journal's scope.

2. Dataset and Data Collection

The dataset used in this project consists of journal articles related to Natural Language Processing (NLP), collected through the **Semantic Scholar API**. The study focused on publications from **2012 to 2025**, enabling an analysis of thematic alignment and research evolution over time.

The collected dataset included **874 research papers**, along with associated metadata such as **titles, publication years, and abstracts**. Since abstracts summarize the key objectives and contributions of research papers, they were selected as the primary textual source for semantic analysis.

Before analysis, the dataset underwent basic preprocessing to ensure consistency and usability. Entries with missing abstracts or incomplete information were removed, and publication years were converted into a structured format for temporal analysis. The cleaned dataset was then organized to support yearly trend analysis, semantic similarity computation, and topic exploration.

The final dataset formed the basis for all subsequent stages of the project, including embedding generation, alignment scoring, semantic visualization, and topic analysis.

3. Methodology

This project applies Natural Language Processing (NLP) techniques to evaluate how closely journal articles align with the thematic focus of the journal. The methodology combines semantic embeddings, similarity measurements, topic analysis, and visualization techniques to analyze research patterns across time.

3.1 Text Representation Using Sentence-BERT

To capture the semantic meaning of research abstracts, this study used **Sentence-BERT (SBERT)**, specifically the **MiniLM variant**, to generate dense vector representations of text. Unlike traditional keyword-based approaches, Sentence-BERT represents text in a semantic embedding space, allowing meaning and contextual similarity to be captured more effectively.

Each article abstract was converted into a numerical vector representation (embedding). In addition, a reference journal scope description was created to represent the thematic objectives of the journal. These embeddings served as the foundation for similarity measurement.

3.2 Alignment Scoring Using Cosine Similarity

To evaluate thematic alignment, **cosine similarity** was used to measure the relationship between article embeddings and the journal scope embedding. Cosine similarity measures the angular similarity between two vectors and produces a score between **0 and 1**, where higher values indicate stronger semantic similarity.

The similarity between each paper abstract and the journal scope was calculated to produce an **alignment score**. These scores were later used to analyze global consistency, yearly variation, and thematic outliers within the dataset.

Similarity(A,B)= (A·B) / (||A|| ||B||)

3.3 Topic Analysis

To understand the thematic composition of the journal, topic analysis was performed using TF-IDF keyword extraction and semantic grouping techniques. Frequently occurring keywords were identified and grouped into broader categories such as **NLP Core, Machine Learning, Language Models, Applications, and Evaluation**.

This approach enabled the comparison of thematic alignment across research areas and supported the analysis of topic evolution over time.

3.4 Semantic Visualization Using PCA

To visualize semantic relationships between papers, **Principal Component Analysis (PCA)** was applied to reduce high-dimensional embeddings into a two-dimensional representation.

PCA visualization allowed semantically similar papers to appear closer together in a shared space, making it possible to identify thematic clusters and outlier papers. Additional visualizations were created by coloring papers according to alignment score to examine how semantic positioning relates to thematic consistency.

4. Results and Analysis

This section presents the findings obtained from semantic alignment scoring, publication trend analysis, semantic visualization, and topic exploration. The results provide insight into how consistently journal articles align with the thematic scope of the journal and how research trends evolve over time.

4.1 Global Thematic Alignment

The distribution of alignment scores revealed a relatively stable thematic consistency across the dataset. Most research papers exhibited alignment scores centered around **0.34–0.35**, indicating moderate to strong similarity with the journal's thematic focus.

The close relationship between the **mean and median alignment scores** suggests a balanced distribution with limited skewness, while the relatively low standard deviation indicates low variability among papers. These findings suggest that the majority of published research remains semantically consistent with the journal's intended scope.

Additionally, yearly analysis demonstrated that alignment remained relatively stable over time, generally ranging between **0.32 and 0.37**. Although a noticeable decline was observed around **2014**, overall trends indicate that the journal maintains a consistent thematic identity across years.

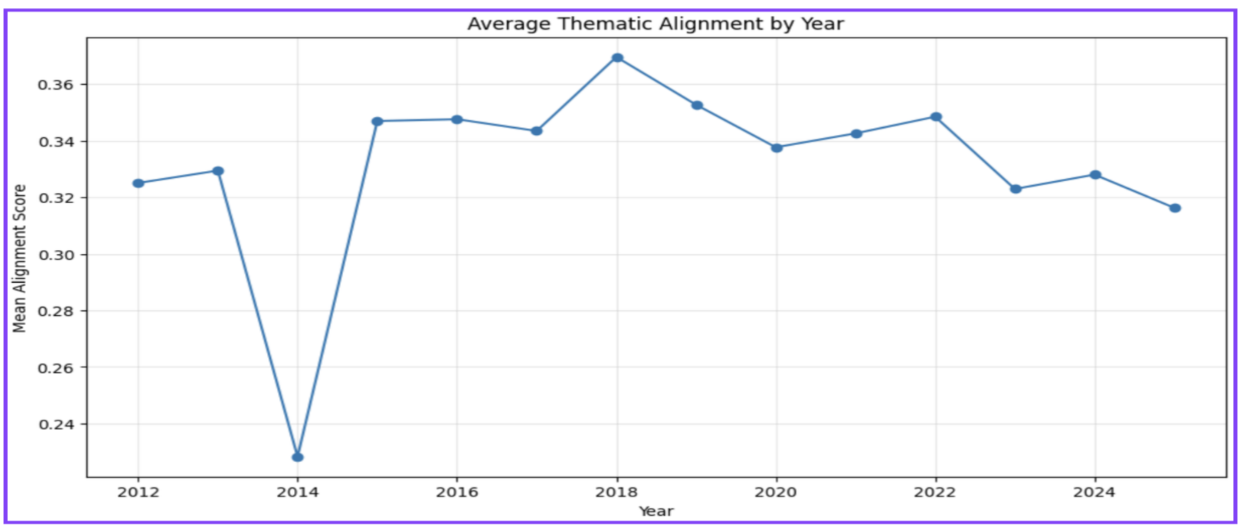

Figure 1: Average Thematic Alignment by Year

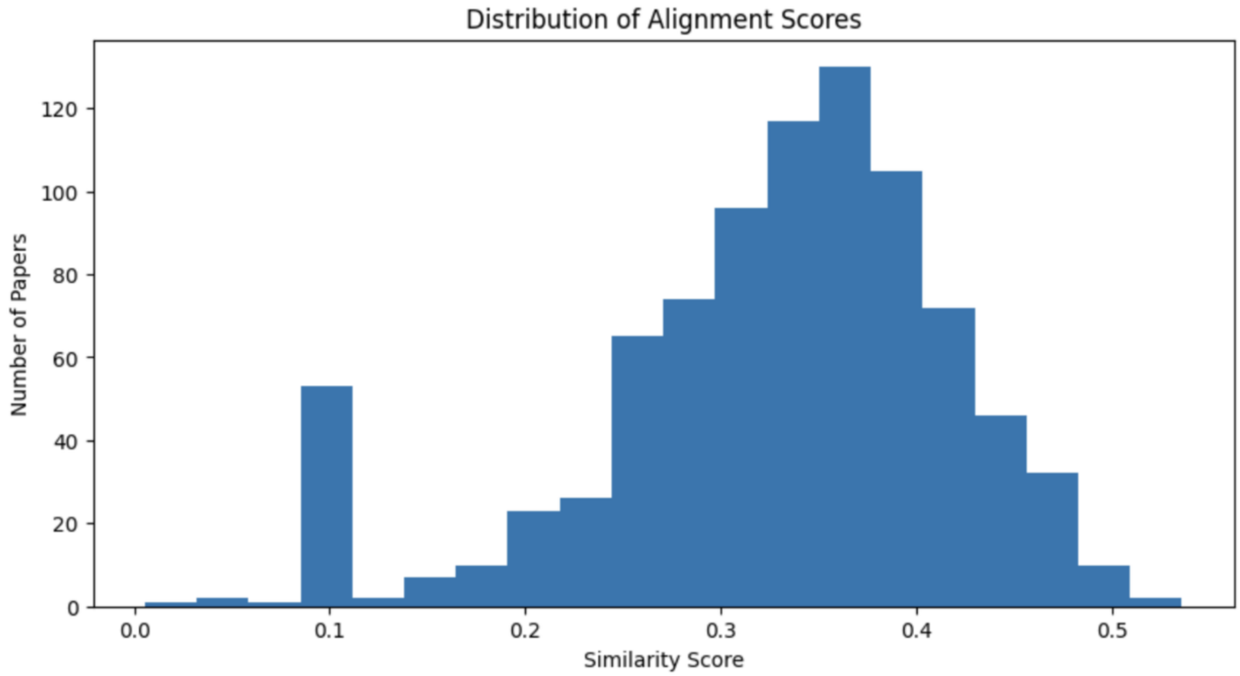

Figure 2: Distribution of Alignment Scores

4.2 Publication Trends Over Time

The analysis of publication volume revealed a clear upward trend in the number of papers published over time. Research activity increased significantly after **2020**, with peak publication volume occurring between **2021 and 2024**.

This increase may reflect the rapid expansion of Natural Language Processing research, particularly due to advances in language models, machine learning techniques, and growing interest in AI-driven language technologies. A slight reduction in publication volume in **2025** is likely influenced by incomplete publication coverage rather than an actual decline in research output.

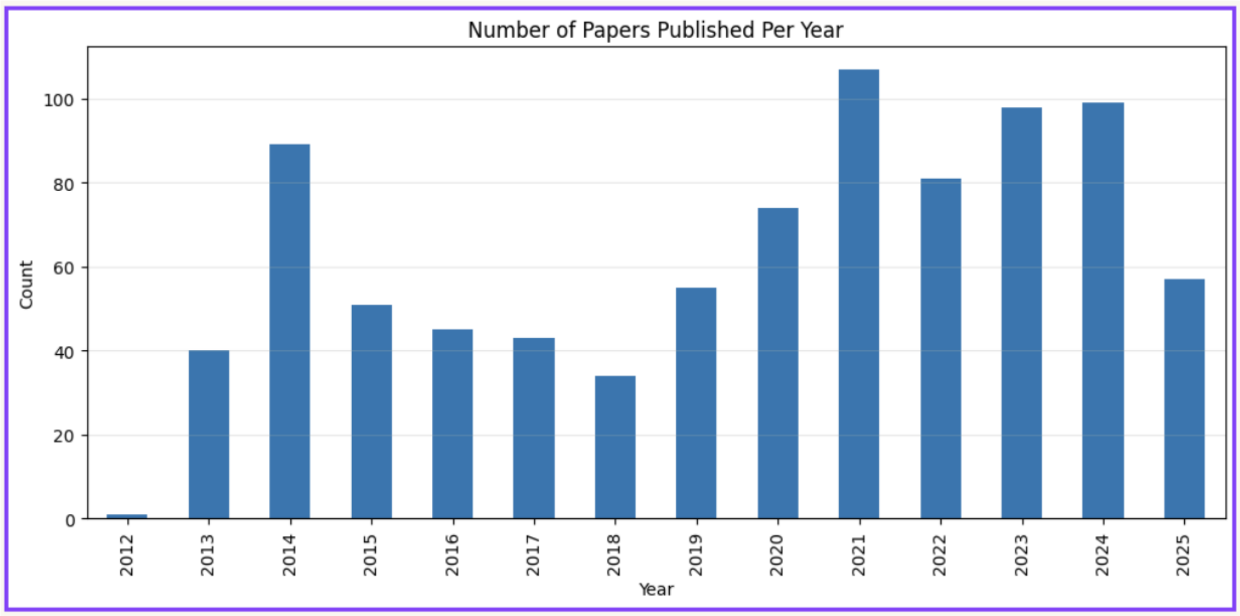

Figure 3: Publication Volume Over Time

4.3 Outlier Analysis

To further understand thematic consistency, papers with the highest and lowest alignment scores were examined. High-alignment papers demonstrated strong semantic similarity with the journal scope and were generally associated with core NLP research areas.

In contrast, low-alignment papers showed weaker semantic similarity and may represent interdisciplinary, niche, or less central research directions. While most papers remained closely aligned with the journal's focus, the presence of outliers highlights natural thematic diversity within published research.

4.4 Semantic Distribution of Papers

To visualize semantic relationships between research papers, PCA-based dimensionality reduction was applied to article embeddings. The resulting semantic distribution showed that most papers formed a dense central cluster, indicating strong thematic similarity across publications.

A smaller number of papers appeared farther from the main cluster, suggesting the presence of thematic outliers. When alignment scores were incorporated into the visualization, highly aligned papers tended to concentrate within core semantic regions, while lower-alignment papers appeared more scattered. This finding supports the validity of the alignment scoring approach and demonstrates consistency between semantic structure and similarity measurement.

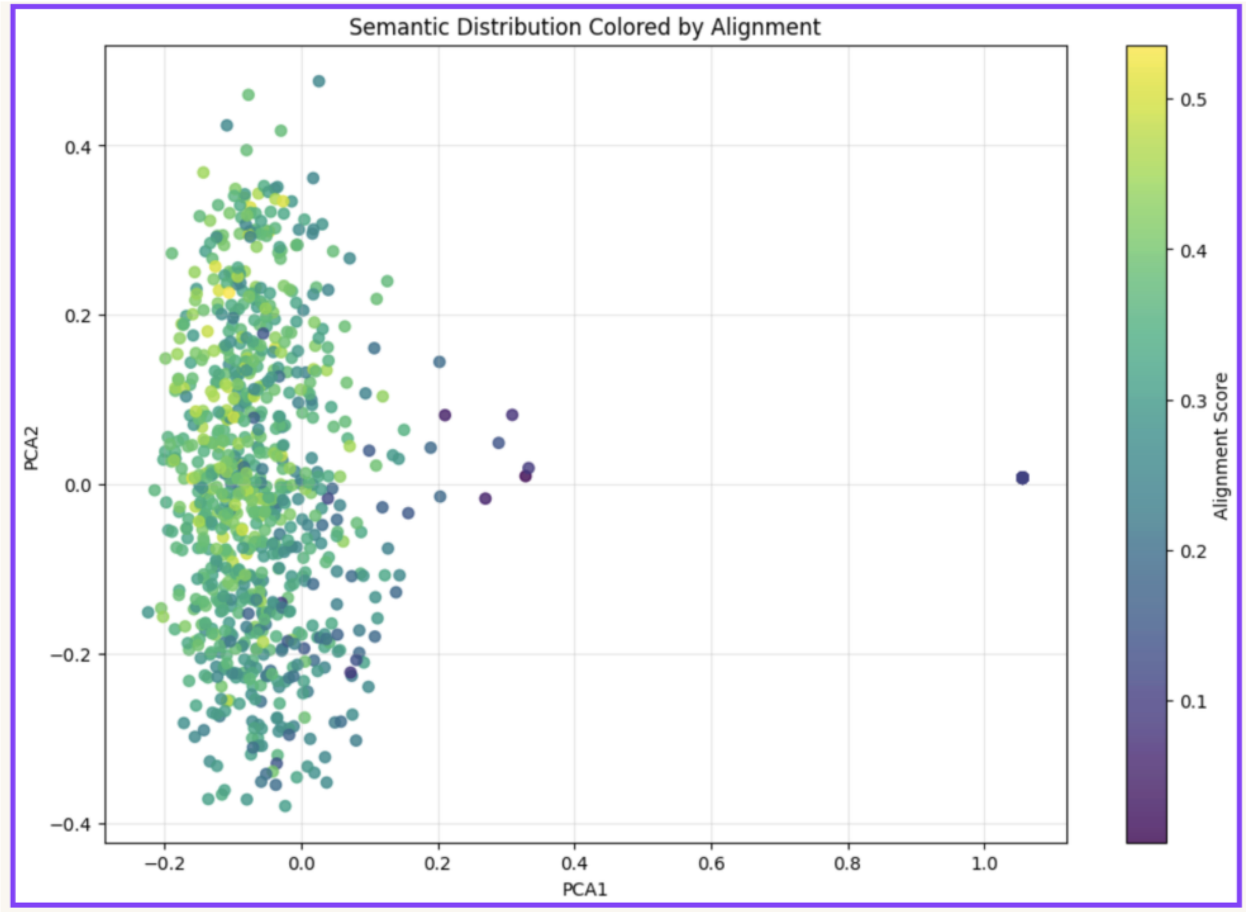

Figure 4: PCA-Based Semantic Distribution of Papers by Alignment Score

4.5 Topic Analysis and Research Evolution

Topic analysis revealed differences in thematic alignment across research areas. **NLP Core** topics showed the highest average alignment scores, while **Evaluation-related research** demonstrated comparatively lower alignment. However, differences across categories remained moderate, suggesting that most research areas remain reasonably aligned with the journal's scope.

Temporal topic analysis further showed that **NLP Core** remained the dominant research area throughout the years. At the same time, **Language Models** and **Machine Learning** displayed strong growth after **2020**, reflecting broader developments within the NLP research community. These findings indicate that while the journal maintains its thematic foundation, research priorities continue to evolve alongside advances in the field.

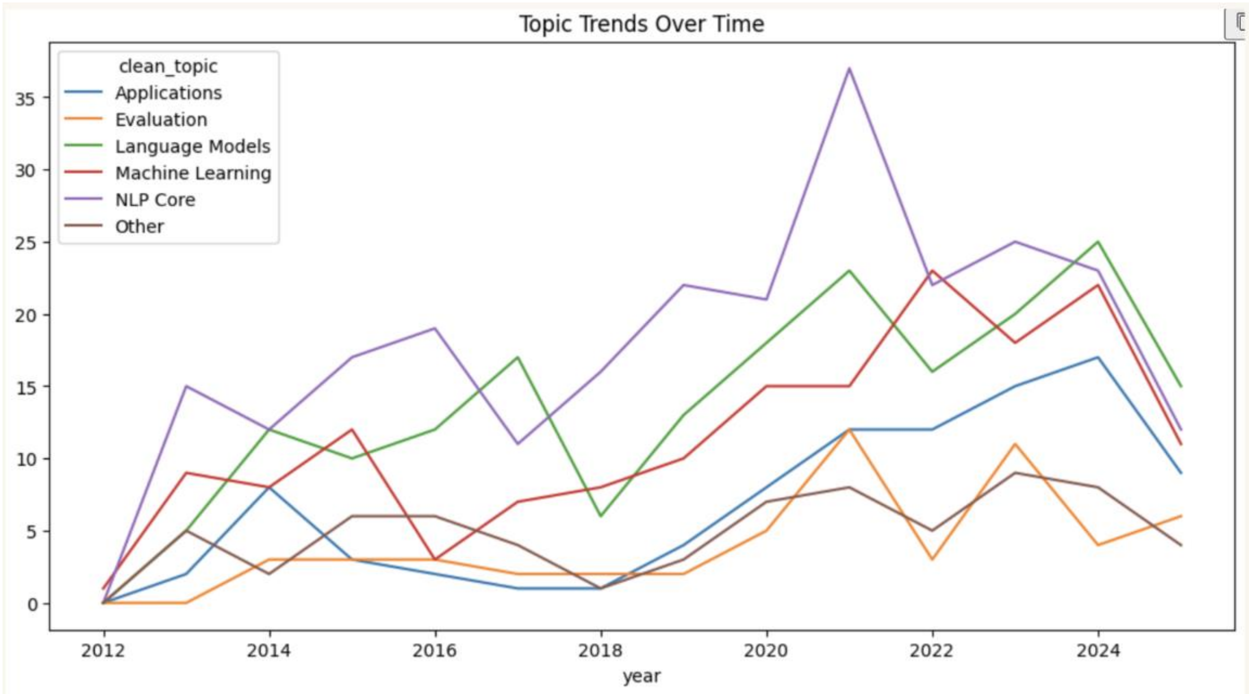

Figure 5 : Evolution of Research Topics Over Time

5. Limitations

Although the proposed methodology provides useful insights into thematic alignment, several limitations should be acknowledged.

First, the analysis is based primarily on **paper abstracts rather than full-text articles**. While abstracts summarize key contributions, they may not fully capture the complete thematic content of research papers.

Second, thematic alignment was measured using **semantic embedding similarity**, which depends on the quality and representational power of the embedding model. Although Sentence-BERT captures semantic meaning effectively, certain contextual nuances or domain-specific relationships may not be fully represented.

Additionally, topic categorization involved a degree of **manual interpretation and grouping**, which may introduce minor subjectivity in how broader research themes are defined.

Finally, publication data for **2025 may be incomplete**, meaning recent trends should be interpreted cautiously.

Despite these limitations, the methodology provides a reliable and data-driven approach for analyzing thematic consistency and research evolution in NLP journal publications.

6. Conclusion

This project investigated thematic alignment in NLP journal articles using semantic embeddings, similarity analysis, and topic exploration techniques.

By collecting and analyzing **874 journal articles**, the study demonstrated that thematic alignment remains relatively stable over time, indicating strong consistency between published papers and the journal's intended focus. Global alignment analysis revealed balanced score distributions with limited variability across publications.

Publication trend analysis further highlighted significant growth in research activity after **2020**, reflecting the rapid expansion of NLP and AI-related research. Semantic visualization through PCA showed that most papers occupy a shared semantic space, while a limited number of outliers indicate thematic diversity.

Topic analysis revealed that **NLP Core** topics maintain the strongest presence throughout the years, while **Language Models** and **Machine Learning** have gained increasing importance in recent years.

Overall, the findings suggest that the journal maintains a **strong and evolving NLP research identity**, balancing thematic consistency with emerging research directions.

7. Future Work

Several opportunities exist to extend this work in future research.

Future studies could analyze **full-text papers instead of abstracts**, enabling deeper semantic understanding of research contributions. Additional comparisons across **multiple NLP journals** may provide broader insight into thematic differences and publication focus.

More advanced topic modeling approaches could also be explored to identify finer-grained research themes and emerging directions. Furthermore, incorporating **citation networks and bibliometric analysis** may provide a richer understanding of how research communities evolve over time.

8. References

- Grootendorst, M. (2022). *BERTopic: Neural topic modeling with a class-based TF-IDF procedure*.
- Semantic Scholar API Documentation.
- Manning, C., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*.